Factorising, substitution and systems

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L2-AL1 Level 2 Intermediate · Algebra

Factorising, the difference of two squares, substitution and combining equations, so that the structure of an expression replaces long arithmetic.
By the end you can spot a common factor or a difference of two squares before expanding anything; handle symmetric problems by working with the sum and product instead of the unknowns; solve contest systems and integer equations by adding, multiplying or factorising.
Prerequisites: Thinking with algebra (L1-AL1). **Contests:** IMC, Intermediate Kangaroo, Cayley, Hamilton, AMC 10. **Time:** about 75 minutes.

1 Lesson

Evaluate

$$\frac{2026^3 - 2026}{2025 \times 2027}.$$

You could multiply everything out, but the cube of 2026 has ten digits and a long division would follow. Instead, note that both terms of the numerator contain a factor of 2026, so $2026^3 - 2026 = 2026\left(2026^2 - 1\right)$. The bracket is a square minus one, which factorises as $(2026-1)(2026+1) = 2025 \times 2027$. This is the denominator, so it cancels and the answer is 2026.

Contest algebra questions are usually built around a factorisation or a substitution, and finding it is quicker and less error-prone than expanding. This unit covers four techniques: factorising, the difference of two squares, substitution, and combining equations before solving them.

1.1 Factorise, don't expand

Expanding is mechanical. Factorising gives information: a product shows divisibility, shows when an expression is zero, and shows what cancels. You should know these identities by heart.

Core identities.
For all numbers a, b, x, y:

$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a-b)^2 = a^2 - 2ab + b^2, \qquad a^2 - b^2 = (a-b)(a+b),$$

$$xy + x + y + 1 = (x+1)(y+1).$$

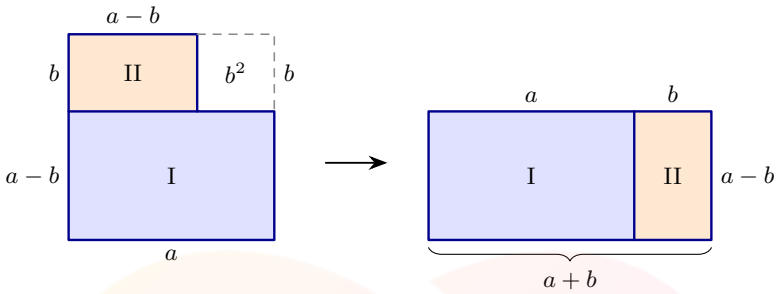
Each one is checked by expanding the right-hand side. The last one is less familiar, but it solves a whole family of integer equations. If you see xy, x and y together, try to complete a product by adding a constant. For instance $(x-2)(y-3) = xy - 3x - 2y + 6$, so $xy - 3x - 2y$ is the same as $(x-2)(y-3) - 6$.

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Before expanding anything, check for a common factor, a perfect square or a difference of squares.

1.2 The difference of two squares

The identity $a^2 - b^2 = (a-b)(a+b)$ gets its own section, because it is used with numbers as often as with letters. The picture shows why it is true. Remove a $b \times b$ square from the corner of an $a \times a$ square, cut the L-shape that is left into two rectangles, and slide one of them round to make a single rectangle of width $a+b$ and height $a-b$.



It is used in three ways.

- Arithmetic: $97 \times 103 = 100^2 - 3^2 = 9991$, and $2025^2 - 2024^2 = 2025 + 2024 = 4049$.
- Square roots: $\left(\sqrt{m} - \sqrt{n}\right)\left(\sqrt{m} + \sqrt{n}\right) = m - n$, so knowing $\sqrt{m} - \sqrt{n}$ tells you $\sqrt{m} + \sqrt{n}$ as well.
- Integers: to solve $a^2 - b^2 = N$ in integers, factorise N.

The third use gives the following theorem.

Which integers are differences of two squares?.
An integer N can be written as $a^2 - b^2$ with a and b integers if and only if N is odd or a multiple of 4.

Proof. The numbers $a-b$ and $a+b$ differ by $2b$, which is even, so they are both odd or both even. If both are odd, then $N = (a-b)(a+b)$ is odd. If both are even, then N is a multiple of $2 \times 2 = 4$. So a number that is 2 more than a multiple of 4, such as 6 or 10, is never a difference of two squares. Conversely, every odd number $2k+1$ equals $(k+1)^2 - k^2$, and every multiple of four $4k$ equals $(k+1)^2 - (k-1)^2$. □

1.3 Substitution

If the same expression keeps appearing, give it a name. The equation $\left(x^2+x\right)^2 - 8\left(x^2+x\right) + 12 = 0$ looks like a quartic, but with $t = x^2 + x$ it reads $t^2 - 8t + 12 = 0$, so $t = 2$ or $t = 6$. Hence $x^2 + x = 2$ gives $x = 1$ or $x = -2$, and $x^2 + x = 6$ gives $x = 2$ or $x = -3$. So the quartic reduces to two quadratics.

The most important case of substitution is symmetry. When a problem treats x and y in the same way, work with the sum $s = x + y$ and the product $p = xy$ rather than with x and y themselves.

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Symmetric expressions.
If $s = x + y$ and $p = xy$, then

$$x^2 + y^2 = s^2 - 2p, \qquad x^3 + y^3 = s^3 - 3ps, \qquad \frac{1}{x} + \frac{1}{y} = \frac{s}{p} \quad (p \neq 0).$$

Conversely, given s and p, the numbers x and y are the two roots of $t^2 - st + p = 0$, so real x and y exist exactly when $s^2 \geq 4p$.

The first identity is $(x+y)^2$ expanded. For the second, $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 = x^3 + y^3 + 3xy(x+y)$. For the third, put the fractions over the common denominator xy. For the last statement, $(t-x)(t-y) = t^2 - st + p$, and a quadratic has real roots exactly when its discriminant, here $s^2 - 4p$, is not negative.

Note that we often never need x and y themselves. If $x + y = 4$ and $xy = 1$, then $x^2 + y^2 = 16 - 2 = 14$ at once, whereas finding $x = 2 \pm \sqrt{3}$ and squaring is slower.

1.4 Simultaneous equations: combine before you substitute

Substituting one equation into another always works in principle, but contest systems are usually set so that combining the equations first is much shorter. We have the following strategies.

- **Add everything.** If each unknown appears equally often, adding gives the total. From $a + b = 9$, $b + c = 13$ and $c + a = 14$ we get $2(a + b + c) = 36$, so $a + b + c = 18$, and then $c = 18 - 9 = 9$, $a = 5$ and $b = 4$.
- **Subtract to cancel.** Subtracting two similar equations often leaves something that factorises.
- **Multiply everything.** When the equations are products, multiplying them together plays the role that adding plays for sums.

Also read what the question actually asks for. It is often a combination such as $x + y$ or xy, which can be found without solving for each unknown.

Example 1.
Find all pairs of positive integers (a, b) with $a^2 - b^2 = 77$.

Solution.
Listing squares, we find $81 - 4 = 77$, so the pair $(9, 2)$ works. A search like this does not tell us when to stop or whether a pair was missed, so we factorise instead.
Write the equation as $(a-b)(a+b) = 77$. As a and b are positive, $a + b$ is positive, so $a - b$ is positive as well, and $a - b$ is smaller than $a + b$. The factor pairs of 77 are 1×77 and 7×11. From $a - b = 1$ and $a + b = 77$ we get $a = 39$, $b = 38$. From $a - b = 7$ and $a + b = 11$ we get $a = 9$, $b = 2$.
So there are exactly two pairs, $(39, 38)$ and $(9, 2)$. Had N been even, we would also have needed $a - b$ and $a + b$ to have the same parity, which rules out some factor pairs.

Example 2.
The real number x satisfies $x + \frac{1}{x} = 3$. Find $x^2 + \frac{1}{x^2}$ and $x^3 + \frac{1}{x^3}$.

Solution.
You could solve for x: multiplying by x gives $x^2 - 3x + 1 = 0$, so $x = \frac{3 \pm \sqrt{5}}{2}$. Cubing this is

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slow. Instead, note that x and $\frac{1}{x}$ have sum 3 and product 1. This is the symmetric situation with $s = 3$ and $p = 1$.
Squaring, $\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$, so $x^2 + \frac{1}{x^2} = 9 - 2 = 7$. Cubing,

$$\left(x + \frac{1}{x}\right)^3 = x^3 + 3x + \frac{3}{x} + \frac{1}{x^3} = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right),$$

so $x^3 + \frac{1}{x^3} = 27 - 9 = 18$.
When you square or cube a number plus its reciprocal, the cross terms are constants or multiples of what you already know. Squaring 7 the same way gives $x^4 + \frac{1}{x^4} = 49 - 2 = 47$, still without ever finding x.

Example 3.
Positive real numbers x, y and z satisfy $x^2yz = 60$, $xy^2z = 90$ and $xyz^2 = 150$. Find $x + y + z$.

Solution.
Substitution is possible but slow: from the first two equations $\frac{y}{x} = \frac{90}{60} = \frac{3}{2}$, from the last two $\frac{z}{y} = \frac{150}{90} = \frac{5}{3}$, and then the first equation becomes an equation in x alone. The following method is shorter and also works for larger systems.
Each unknown appears in all three equations, twice in one of them and once in each of the others, so multiply all three: $(xyz)^4 = 60 \times 90 \times 150 = 810000 = 30^4$. Everything is positive, so $xyz = 30$. Each equation is xyz times one of the unknowns, so dividing by xyz gives $x = \frac{60}{30} = 2$, $y = \frac{90}{30} = 3$ and $z = \frac{150}{30} = 5$. So $x + y + z = 10$.
This is the multiplicative version of adding all the equations. Note that positivity was used: for real numbers, $(xyz)^4 = 30^4$ also allows $xyz = -30$, and then $x = -2$, $y = -3$, $z = -5$ satisfy all three equations, giving $x + y + z = -10$.

Example 4.
How many pairs of positive integers (x, y) satisfy $xy - 2x - 3y = 6$?

Solution.
We could make x the subject: $x(y - 2) = 3y + 6$, so $x = \frac{3y+6}{y-2}$. Trying $y = 3, 4, 5, \ldots$ produces some solutions, but again we do not know when to stop. The combination xy, $-2x$, $-3y$ suggests adding a constant to complete a product.
We want $(x - 3)(y - 2)$, which expands to $xy - 2x - 3y + 6$. So add 6 to both sides: $(x - 3)(y - 2) = 12$. Now $x - 3$ and $y - 2$ are integers with product 12. Could both be negative? Since $x \geq 1$ and $y \geq 1$, we would have $x - 3 \geq -2$ and $y - 2 \geq -1$, and two negative integers of that size have product at most 2. So both factors are positive, and each positive divisor d of 12 gives exactly one solution, $x = 3 + d$ and $y = 2 + \frac{12}{d}$.
As 12 has the six positive divisors $1, 2, 3, 4, 6, 12$, there are 6 pairs: $(4, 14)$, $(5, 8)$, $(6, 6)$, $(7, 5)$, $(9, 4)$ and $(15, 3)$. The first attempt was not wrong: $\frac{3y+6}{y-2} = 3 + \frac{12}{y-2}$ says the same thing. Either way, the factorisation is what shows that the list is complete.

Common traps.

- $(a + b)^2$ is not $a^2 + b^2$. The cross term $2ab$ is usually what the question is testing.
- Cancelling terms instead of factors. $\frac{a^2+b^2}{a+b}$ does not simplify: only a factor of the whole top and the whole bottom can cancel.

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- Dividing by something that might be zero. From $x^2 = 5x$ you cannot conclude $x = 5$. Factorise to $x(x-5) = 0$ and keep both roots.
- In integer factor problems, forgetting the negative factor pairs, or forgetting that $a - b$ and $a + b$ must have the same parity.
- Finding s and p but never checking that real x and y exist, which needs $s^2 \geq 4p$.

1.5 How to spot this in a contest

Large numbers sitting next to each other (like 2025, 2026, 2027) or close to a round number nearly always mean a factorisation: call the middle one n and write the others as $n \pm 1$. A question that gives you $x + y$ and xy and asks for $x^2 + y^2$, $x^3 + y^3$ or $\frac{1}{x} + \frac{1}{y}$ is asking you to work with the sum and product. An equation in positive integers that mixes xy with x and y needs a constant added to complete a product. Three or more equations with a cyclic pattern should be added or multiplied. A repeated expression should be given a name.

Whenever you are about to expand something large, first look for a common factor or a difference of squares. On the IMC and the AMC 10 this often turns a ten-minute question into a two-minute one.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Evaluate $\frac{2024^2 - 576}{2000}$.

Problem 2 ⋆⋆ Find the sum of the squares of all the real solutions of

$$\left(x^2 - 5x\right)^2 - 2\left(x^2 - 5x\right) - 24 = 0.$$

Problem 3 ⋆⋆ Real numbers a and b satisfy $a + b = 6$ and $ab = 4$. Find $a^3 + b^3$.

Problem 4 ⋆⋆⋆ What is the smallest positive integer that can be written in the form $a^2 - b^2$, where a and b are positive integers, in three different ways?

Problem 5 ⋆⋆⋆ Positive integers x and y satisfy $xy = 2x + 5y + 10$. Find the sum of all possible values of x.

Problem 6 ⋆⋆⋆⋆ Real numbers x and y satisfy

$$x^2 + xy + y^2 = 19 \qquad \text{and} \qquad x + y + xy = 11.$$

Find $x^3 + y^3$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

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1. **AMC 10A 2016 Problem 1** ⋆
   The factorials share a common factor
2. **UKMT IMC 2018 Question 10** ⋆
   Think about the parity of the two factors
3. **UKMT IMC 2017 Question 12** ⋆⋆
   Combine the given totals before finding anyone's age
4. **AMC 10A 2017 Problem 5** ⋆⋆
   Write the sum of reciprocals over a common denominator
5. **AMC 10B 2019 Problem 6** ⋆⋆
   Divide through by the smallest factorial
6. **AMC 10A 2018 Problem 10** ⋆⋆
   Multiply the two square roots by their conjugate
7. **AMC 10A 2020 Problem 14** ⋆⋆⋆
   Factorise first, then use the sum and product
8. **UKMT IMC 2017 Question 16** ⋆⋆⋆
   Use one equality to relate the two integers, then substitute
9. **AMC 10A 2021 Problem 9** ⋆⋆⋆
   Expand, then look for a different factorisation
10. **AMC 10A 2021 Problem 10** ⋆⋆⋆
    Multiply by a factor that starts a chain of differences of squares
11. **AMC 10A 2019 Problem 19** ⋆⋆⋆⋆
    Pair the four brackets so that the two products share a common expression
12. **AMC 10A 2020 Problem 21** ⋆⋆⋆⋆
    Name 2^{17} and think about dividing a sum of odd powers by $a+1$

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4 Hints

1. Is 576 a perfect square?
2. Give the repeated expression a name and solve the quadratic you get first.
3. Expand $(a+b)^3$ and see which part of it you already know.
4. Write $N = (a-b)(a+b)$. Each way corresponds to a factorisation $N = uv$ with u smaller than v and both of the same parity. Treat odd N and multiples of 4 separately.
5. Move everything to one side and add a constant so that the left-hand side becomes a product of two brackets.
6. Both equations are symmetric. Rewrite them in terms of $s = x+y$ and $p = xy$, and check in each case that real x and y exist.

5 Solutions

Problem 1

Evaluate $\frac{2024^2-576}{2000}$.

Since $576 = 24^2$, the numerator is a difference of two squares: $2024^2 - 24^2 = (2024-24)(2024+24) = 2000 \times 2048$. Dividing by 2000 leaves 2048.

Answer: 2048

Problem 2

Find the sum of the squares of all the real solutions of

$$\left(x^2-5x\right)^2 - 2\left(x^2-5x\right) - 24 = 0.$$

Let $t = x^2 - 5x$. The equation becomes $t^2 - 2t - 24 = 0$, which factorises as $(t-6)(t+4) = 0$, so $t = 6$ or $t = -4$. If $t = 6$, then $x^2 - 5x - 6 = 0$, that is $(x-6)(x+1) = 0$, giving $x = 6$ or $x = -1$. If $t = -4$, then $x^2 - 5x + 4 = 0$, that is $(x-1)(x-4) = 0$, giving $x = 1$ or $x = 4$. These four real solutions are distinct, and the sum of their squares is $36 + 1 + 1 + 16 = 54$.

Answer: 54

Problem 3

Real numbers a and b satisfy $a+b = 6$ and $ab = 4$. Find $a^3 + b^3$.

Expanding, $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$. So $a^3 + b^3 = (a+b)^3 - 3ab(a+b) = 216 - 3 \times 4 \times 6 = 216 - 72 = 144$. Such real numbers do exist, because $(a+b)^2 = 36$ is at least $4ab = 16$; they are $3 \pm \sqrt{5}$.

Answer: 144

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Problem 4

What is the smallest positive integer that can be written in the form $a^2 - b^2$, where a and b are positive integers, in three different ways?

Write $N = uv$ with $u = a - b$ and $v = a + b$. Then u and v are positive with the same parity, and $b \geq 1$ means $v - u = 2b \geq 2$, so u is smaller than v. Conversely, if $N = uv$ with u smaller than v and both of the same parity, then $a = \frac{u+v}{2}$ and $b = \frac{v-u}{2}$ are positive integers with $a^2 - b^2 = N$. Different factorisations give different pairs (a, b), so we need N to have three factorisations of this kind.

If N is 2 more than a multiple of 4 there are none. If N is odd, every factorisation $N = uv$ has both factors odd, so we need three factor pairs with u smaller than v, and hence at least six divisors (for a square, the pair $u = v$ is not allowed, so it needs at least seven). Every odd number below 45 is 1, a prime, the square of a prime (9, 25), the cube 27 or a product of two different primes (15, 21, 33, 35, 39), and so has at most four divisors. The odd number $45 = 3^2 \times 5$ has six. If $N = 4k$, both factors must be even, so $u = 2u'$ and $v = 2v'$ with $u'v' = k$ and u' smaller than v'; we again need k to have at least six divisors, and the smallest such k is 12, giving $N = 48$, which is larger than 45. So the answer is 45.

Its three factorisations 1×45, 3×15 and 5×9 give $45 = 23^2 - 22^2 = 9^2 - 6^2 = 7^2 - 2^2$.

Answer: 45

Problem 5

Positive integers x and y satisfy $xy = 2x + 5y + 10$. Find the sum of all possible values of x.

Rearranging gives $xy - 2x - 5y = 10$. Since $(x - 5)(y - 2) = xy - 2x - 5y + 10$, adding 10 to both sides gives $(x - 5)(y - 2) = 20$. As x and y are positive, $x - 5 \geq -4$ and $y - 2 \geq -1$. If both factors were negative, we would need $y - 2 = -1$ and so $x - 5 = -20$, which is impossible. So both factors are positive, and $x - 5$ is a positive divisor of 20, namely $1, 2, 4, 5, 10$ or 20. Each gives a valid y, and the possible values of x are $6, 7, 9, 10, 15, 25$, whose sum is 72.

Answer: 72

Problem 6

Real numbers x and y satisfy

$$x^2 + xy + y^2 = 19 \qquad \textit{and} \qquad x + y + xy = 11.$$

Find $x^3 + y^3$.

Let $s = x + y$ and $p = xy$. Since $x^2 + xy + y^2 = (x + y)^2 - xy$, the equations become $s^2 - p = 19$ and $s + p = 11$. Adding them gives $s^2 + s = 30$, that is $(s - 5)(s + 6) = 0$. Hence either $s = 5$ and $p = 6$, or $s = -6$ and $p = 17$.

Real x and y with sum s and product p exist only when the quadratic $t^2 - st + p = 0$ has real roots, that is, when $s^2 \geq 4p$. In the first case $25 \geq 24$, so it occurs, with $\{x, y\} = \{2, 3\}$. In the second case 36 is less than 68, so x and y would not be real, and this case is rejected. (It would give $s^3 - 3ps = -216 + 306 = 90$, which is the trap answer.)

So $x^3 + y^3 = s^3 - 3ps = 125 - 90 = 35$, and indeed $2^3 + 3^3 = 35$.

Answer: 35

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The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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